University of Essex

CE205
Databases and Information Retrieval

Multivalued and Join Dependencies, 4NF and 5NF

Dr. Zulfiqar Ali

Lecturer

School of Computer Science and Electronic Engineering

Most of the slides based on

- *Database Systems: A Practical Approach to Design, Implementation, and Management* by Conoly and Begg, Global Edition/6th Edition.
- *Database processing: fundamentals, design, and implementation* by David M. Kroenke; David J. Aue,14th Edition (Global Edition).

Learning Objectives

- Normalisation of a relation straight to BCNF
- Surrogate key
- Foreign Key and Referential Integrity Constraint
- What is multivalued dependency
- Updated anomalies due to multivalued dependencies
- What is fourth normal form (4NF)
- What is Join Dependency and 5NF

AD CAMPAING MIX with Before Normalization (with Update Anomalies)

AD CAMPAIGN MIX

AdCampaignID	AdCampaign Name	StartDate	Duration	Campaign MgrID	Campaign MgrName	ModusID	Media	Range	Budget Pctg
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	1	TV	Local	50%
111	SummerFun20	6.6.2020	12 days	CM100	Roberta	2	TV	National	50%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	1	TV	Local	60%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	3	Radio	Local	30%
222	SummerZing20	6.8.2020	**30 days**	CM101	Sue	5	Print	Local	10%
333	FallBall20	6.9.2020	12 days	CM102	John	3	Radio	Local	80%
333	FallBall20	6.9.2020	12 days	CM102	John	4	Radio	National	20%
444	**AutmnStyle20**	**6.9.2020**	**5 days**	CM103	Nancy	6	Print	National	100%
555	AutmnColors20	6.9.2020	3 days	CM100	Roberta	3	Radio	Local	100%
????	????	????	????	????	????	**7**	**Internet**	**National**	**????**

Deletion Anomaly Example:
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign modus 6.

Modification Anomaly Example:
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified.

Insertion Anomaly Example:
Cannot insert new campaign modus 7 without inserting an actual campaign using the new modus 7.

University of Essex

AD CAMPAING MIX After Normalisation to BCNF

AD CAMPAIGN

AdCampaignID	AdCampaignName	StartDate	Duration	CampaignMgrID
111	SummerFun20	6.6.2020	12 days	CM100
222	SummerZing20	6.8.2020	45 days	CM101
333	FallBall20	6.9.2020	12 days	CM102
555	AutmnColors20	6.9.2020	3 days	CM100

Modification Anomaly Resolved:
Only one record modified.

CAMPAIGN MANAGER

CampaignMgrID	CampaignMgrName
CM100	Roberta
CM101	Sue
CM102	John
CM103	Nancy

MODUS

ModusID	Media	Range
1	TV	Local
2	TV	National
3	Radio	Local
4	Radio	National
5	Print	Local
6	Print	National
7	Internet	National

Deletion Anomaly Resolved:
Campaign 444 deleted but all the data about the campaign manager CM103 and the campaign modus 6 remain in the database.

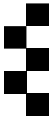
AD CAMPAIGN MIX

AdCampaignID	ModusID	BudgetPctg
111	1	50%
111	2	50%
222	1	60%
222	3	30%
222	5	10%
333	3	80%
333	4	20%
555	3	100%

Insertion Anomaly Resolved:
New campaign modus 7 inserted without inserting an actual campaign using the new modus 7.

University of Essex

Normalise Straight to Boyce-Codd Normal Form (BCNF)



University of Essex

Process to Convert a Relation into BCNF

1. Identify every functional dependency.
2. Identify every candidate key.
3. If there is a functional dependency that has a determinant that is not a candidate key:
   A. Move the columns of that functional dependency into a new relation.
   B. Make the determinant of that functional dependency the primary key of the new relation.
   C. Leave a copy of the determinant as a foreign key in the original relation.
   D. Create a referential integrity constraint between the original relation and the new relation.
4. Repeat step 3 until every determinant of every relation is a candidate key.

Note: In step 3, if there is more than one such functional dependency, start with the one with the most columns.

Scenario

Students' Activities

Design a database to record students' activities and the amount paid by them to each club. A student can join multiple clubs such as football, golf, and chess, and have to pay the fee for each membership.

Example: Straight to BCNF (1)

Sample Data for Relation

STUDENT_ACTIVITY

StudentID	StudentName	Activity	ActivityFee	AmountPaid
100	Jones	Golf	65.00	65.00
100	Jones	Skiing	200.00	0.00
200	Davis	Skiing	200.00	0.00
200	Davis	Swimming	50.00	50.00
300	Garrett	Skiing	200.00	100.00
300	Garrett	Swimming	50.00	50.00
400	Jones	Golf	65.00	65.00
400	Jones	Swimming	50.00	50.00

Example: Straight to BCNF (2)

Now, we have this relation that stores data about student activities:

STUDENT_ACTIVITY (StudentID, StudentName, Activity, ActivityFee, AmountPaid)

where

- **StudentID** is a student identifier,
- **StudentName** is student name,
- **Activity** is the name of a club or other organized student activity,
- **ActivityFee** is the cost of joining the club or participating in the activity, and
- **AmountPaid** is the amount the student has paid toward the ActivityFee.

Example: Straight to BCNF (3)

Determine FDs

StudentID is a unique student identifier, so we know that:

StudentID → StudentName

However, does the functional dependency exist?

StudentID → Activity

- It does if a student belongs to just one club or participates in just one activity, but it does not if a student belongs to more than one club or participates in more than one activity.
- Davis with StudentID 200 participates in both Skiing and Swimming, so StudentID does not determine Club. StudentID does not determine ActivityFee or AmountPaid, either.

Example: Straight to BCNF (4)

Determine FDs

- Now consider the StudentName column.
- Does StudentName determine StudentID? Is, for example, the value 'Jones' always paired with the same value of StudentID?
    - No, there are two students named 'Jones', and they have different StudentID values.
- StudentName does not determine any other column in this table, either.

Example: Straight to BCNF (5)

Determine FDs

Considering the next column, Activity.

- We know that many students can belong to a club. Therefore, Activity does not determine StudentID or StudentName.

- Does Activity determine ActivityFee? Is the value 'Skiing', for example, always paired with the same value of ActivityFee?
    - Activity → ActivityFee

- From these data, it appears so, and using just this sample data, we can conclude that Activity determines ActivityFee.

Example: Straight to BCNF (6)

Determine FDs

So far, we have two functional dependencies:

StudentID → StudentName

Activity → ActivityFee

Are there other functional dependencies with composite determinants?

- No single column determines AmountPaid, so consider possible composite determinants for it.
- AmountPaid is dependent on both the student and the club the student has joined. Therefore, it is determined by the combination of the determinants StudentID and Activity.
- Thus, we can say
- (StudentID, Activity) → AmountPaid

Example: Straight to BCNF (7)

Look for Candidate Keys

- So far, we have three determinants:
  - StudentID, Activity, and (StudentID, Activity).
- Are any of these candidate keys? Do any of these determinants identify a unique row?
  - From the data, it appears that (StudentID, Activity) identifies a unique row and is a **candidate key**.
- In real situations, we would need to check this assumption out with the user.

Example: Straight to BCNF (8)

Check all determinants are candidate keys?

- Relation STUDENT_ACTIVITY is not in BCNF because columns StudentID and Activity are both *determinants* but **neither** is a *candidate key*.

- To normalize this relation, we need to construct sub-relations so that every determinant is a candidate key.
- We can do this by creating a separate table for each functional dependency.

Example: Straight to BCNF (9)

Introduce sub-relations

The result is:

- **PAYMENT** (*StudentID*, *Activity*, AmountPaid)
- **STUDENT** (StudentID, StudentName)
- **ACTIVITY** (Activity, ActivityFee)

with referential integrity constraints:

PAYMENT.StudentID must exist in STUDENT.StudentID

and

PAYMENT.Activity must exist in ACTIVITY.Activity

Straight to BCNF

Consider a relation

R {title, year, studioName, president, presAddr}

which is abbreviated as R (T, Y, S, P, A)

That is, each tuple of this relation tells about a movie, its studio, the president of the studio, and the address of the president of the studio.

Convert relation R to BCNF?